

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Code of Conduct:

Cl. Santosh Kumar 1925240200
I ~~Santosh Kumar~~ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

P. S. S.
Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%				
Method Selection	20%				
Solution Process	25%				
Accuracy of Calculation	20%				
Interpretation of Results	15%				

Total Marks *20* /40

[Signature]
Signature of the Faculty

Solution

To accommodate the different sized subnets without overlap, we allocate them using variable length subnet Masking (VLSM), starting from the largest subnet to the smallest.

Lab A (/26) : Requires $2^{32-26} = 2^6 = 64$ total addresses

Network Address : 192.168.20.0

valid Host Range : 192.168.20.1 to 192.168.20.62

Broadcast Address : 192.168.20.63

usable Hosts : $64 - 2 = 62$

Lab B (/27) : Requires $2^{32-27} = 2^5 = 32$ total addresses.

starts at the ~~next~~ available IP.

Network Address : 192.168.20.64

valid Host Range : 192.168.20.65 to 192.168.20.94

Broadcast Address : 192.168.20.95

usable Host : $32 - 2 = 30$

Lab C(29) : Required $2^{32-28} = 2^4 = 16$ total addresses.
starts at the next available

Network Address : 192.168.20.96

Valid Host Range : 192.168.20.97 to 192.168.20.110

Broadcast Address : 192.168.20.111

Usable Hosts : $16 - 2 = 14$

Solution

The subnet mask 255.255.255.192 corresponds to a /26 prefix. The block size is $256 - 192 = 64$. The subnets fall on multiples of 64: 0, 64, 128 and 192. Since .130 and .190 fall between 128 and 192, both systems belong to the exact same subnet.

Network Address (for both) : 192.168.1.128

Broadcast Address (for both) : 192.168.1.191

Valid Host Range : 192.168.1.129 to 192.168.1.190

Usable Host Addresses : $2^6 - 2 = 62$ usable hosts per subnet.

Network Address : 172.16.10.64

Broadcast Address : 172.16.10.95

IP Address 172.16.10.95

Yes, 172.16.10.95 falls within the exact same subnet range. However, it serves specifically as the broadcast address for the 172.16.10.64/27 subnet. So it cannot be assigned to a standard host.

Solution

Compressed form : Remove leading zeros in each block and replace the longest consecutive block of zeros with a double colon (::).

Result : 2001:db8::ff00:42:8329

Expand (full notation) : Restoring all 16 bits (4 hex digits) per hextet.

Result : 2001:0db8:0000:0000:0000:ff00:0042:8329

network prefix: using the 164 prefix length (the first 64 bits or 4 hexets).

Result: 2001:0db8:0000:0000::1/64
(or) compressed

2001:db8::1/64

Total possible Host Addresses: A /64 network leaves 64 bits for the interface ID (host portion).

Result: 2^{64} host addresses (approx 1.84×10^{19}).